# Weihan Li

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## Education

**Georgia Institute of Technology**, PhD in Machine Learning — Sept 2023 – Present

**Georgia Institute of Technology**, Master of Science in Computer Science — 2023 – 2025

- **Ph.D. Advisor**: Prof. Anqi Wu

**Zhejiang University**, Research Assistant — Sept 2020 – May 2023

- **Advisor**: Prof. Gang Pan

**Zhejiang University**, Bachelor in Computer Science and Technology — Sept 2016 – June 2020

## Research Summary

My current research is in **robot learning for dexterous bimanual manipulation**: how multi-fingered robot hands can learn diverse manipulation skills from human demonstrations, and how skills learned in simulation can transfer reliably to real hardware. Previously, I studied how multiple brain regions communicate and how animal behavior can be understood at scale, with applications to brain disorders and brain-computer interfaces.

## Research Experience: Robot Learning

**Dexterous Bimanual Manipulation from Human Demonstrations** — Aug 2026 – Present

Platform: bimanual 20-DoF dexterous hands (Wuji, Sharpa) on a Vega humanoid

- Building a pipeline that turns human hand–object motion capture into contact-rich bimanual skills for two 20-DoF dexterous hands on 7-DoF arms, aimed at complementing VLAs with reliable dexterous execution.
- Converting demonstrations into physically feasible robot references through kinematic retargeting, collision-aware arm IK, and physics-based retargeting.
- Training residual reinforcement-learning policies in GPU simulation (Newton / MuJoCo Warp, Isaac Lab) with contact-based rewards, robustness to perturbations, and safety constraints.
- Working toward sim-to-real transfer by matching simulation to vendor hardware specifications and evaluating policies against hardware-readiness criteria.

## Research Experience: AI for Neuroscience and Behavior

**Mouse Behavior Analysis** — Sep 2025 – Jul 2026

- **ResiFuse** (NeurIPS 2026; first author): Pose–video fusion that adds video features only on frames with large quantization residuals in a pose-based motion encoder; outperforms pose- and video-based baselines on most tasks across three mouse datasets, with the largest gains in the most socially complex setting.
- **MICE** (NeurIPS 2026 Spotlight): A hierarchical self-supervised foundation model for mouse behavior, pretrained jointly on six mouse-pose datasets; a Perceiver-style social encoder lets a single checkpoint handle any number of animals.
- **QDMouse4M** (NeurIPS 2026 E&D Track): A six-view dataset of over four million frames of freely moving mice, with 2D and 3D keypoint labels from subdermal quantum-dot fluorescence markers, for benchmarking pose estimation and behavior analysis.
- **BehaviorVLM** (preprint; co-first author): A finetuning-free framework for pose estimation and behavioral understanding that guides pretrained vision–language models through explicit, verifiable reasoning steps.

**Multi-Region Brain Communication** — Sep 2023 – May 2026

- **MBRILA** (open-source software; creator): A general Gaussian-process state-space (GP-SSM) framework in which any Gaussian-process model can be converted into an equivalent state-space form with scalable inference; MRM-GP and ADM are built on it. MIT license, `github.com/BRAINML-GT/MBRILA`.
- **ADM** (ICML 2025 **Oral**; first author): An adaptive delay model that combines Gaussian processes with

state-space models to learn time-varying, directional communication across brain regions, with parallel-scan inference that reduces the time complexity from $O(T^3)$ to $O(\log T)$.

- **MRM-GP** (ICML 2024; first author): A linear dynamical system that mirrors a multi-output Gaussian process with explicit frequencies and phase delays, recovering directional, frequency-specific communication across brain regions in linear time.

**Statistical Models of Neural Activity** Jan 2022 – 2026

- **FacRNN** (ICML 2026 Spotlight): A generative low-rank RNN framework that encourages group-wise independence among latent dynamics, yielding more interpretable neural latent trajectories and low-rank connectivity.
- **MIG-Vis** (ICLR 2026): Mutual-information-guided diffusion that visualizes the visual-semantic features encoded by disentangled neural latent groups in macaque IT cortex.
- **BeNeDiff** (NeurIPS 2024): Behavior-relevant, disentangled neural subspaces, with video diffusion models that generate behavior videos revealing what each neural factor encodes.
- **POGLM** (ICML 2024): A differentiable partially observable GLM with forward–backward message passing for learning neural connectivity in the presence of hidden neurons.
- **VIS** (ICLR 2024 Spotlight): Variational importance sampling that maximizes the log-likelihood of latent variable models using a proposal obtained by minimizing the forward $\chi^2$ divergence.
- **HDPP** (NeurIPS 2022; first author): A hierarchical Dirichlet point process that detects neural sequences online with particle filters and introduces new sequence types on the fly.

## Industry Experience

**TikTok**, Video Recommendation May 2026 – Aug 2026

*Research Intern*

- Explored model architectures for large recommendation models that combine multimodal video content representations with long sequences of user behavior features.
- Studied how to jointly model video multimodal signals and long-range user histories to improve video recommendation.

## Publications

***Learning When Visual Context Matters for Mouse Behavior Analysis*** **NeurIPS 2026** [link]
***Weihan Li***, Jingyang Ke, Qinheng Pu, Yule Wang, Chengrui Li, Anqi Wu

***MICE: Multi-animal Interaction Context Encoder — A Hierarchical Foundation Model for Mouse Behavior*** **NeurIPS 2026 Spotlight**
Yen-Shuo Su, ***Weihan Li***, Anqi Wu

***QDMouse4M: A Multi-View 3D Mouse Spontaneous Behavior Dataset with Quantum-Dot Markers*** **NeurIPS 2026 E&D Track**
Jingyang Ke, Amartya Pradhan, Xueling Zhang, ***Weihan Li***, Anqi Wu, Jeffrey E. Markowitz

***BehaviorVLM: Unified Finetuning-Free Behavioral Understanding with Vision-Language Reasoning*** preprint [link]
Jingyang Ke[*], ***Weihan Li***[*], Amartya Pradhan, Jeffrey Markowitz, Anqi Wu
[*]*Equal contribution; authors listed alphabetically.*

***A Factorized Low-Rank RNN Framework for Uncovering Independent Neural Latent Dynamics and Connectivity*** **ICML 2026 Spotlight** [link]
Chengrui Li, Yunmiao Wang, Yule Wang, ***Weihan Li***, Dieter Jaeger, Anqi Wu

***Uncovering Semantic Selectivity of Latent Groups in Higher Visual Cortex with Mutual Information-Guided Diffusion*** **ICLR 2026** [link]
Yule Wang, Joseph Yu, Chengrui Li, ***Weihan Li***, Dieter Jaeger, Anqi Wu

***Learning Time-Varying Multi-Region Brain Communications via Scalable Markovian Gaussian Processes*** **ICML 2025 Oral** [link]
***Weihan Li***, Yule Wang, Chengrui Li, Anqi Wu

***Exploring Behavior-Relevant and Disentangled Neural Dynamics with Generative Diffusion Models*** **NeurIPS 2024** [link]
Yule Wang, Chengrui Li, ***Weihan Li***, Anqi Wu

***Multi-Region Markovian Gaussian Process: An Efficient Method to Discover Directional Communications Across Multiple Brain Regions*** **ICML 2024** [link]
***Weihan Li***, Chengrui Li, Yule Wang, Anqi Wu

***A Differentiable Partially Observable Generalized Linear Model with Forward-Backward Message Passing*** **ICML 2024** [link]
Chengrui Li, ***Weihan Li***, Yule Wang, Anqi Wu

***Forward $\chi^2$ Divergence Based Variational Importance Sampling*** **ICLR 2024 Spotlight** [link]
Chengrui Li, Yule Wang, ***Weihan Li***, Anqi Wu

***Online Neural Sequence Detection with Hierarchical Dirichlet Point Process*** **NeurIPS 2022** [link]
***Weihan Li***, Yu Qi, Gang Pan

***Efficient Point-Process Modeling of Spiking Neurons for Neuroprosthesis*** **EMBC 2021** [link]
***Weihan Li***, Cunle Qian, Yu Qi, Yiwen Wang, Yueming Wang, Gang Pan

## Technical Skills

**Robotics & Simulation**: Isaac Lab, MuJoCo, Motion Retargeting, Sim-to-Real Transfer

**Machine Learning**: PyTorch, Reinforcement Learning, Probabilistic Modeling, Vision-Language Models

**Languages**: Python, C++

## Teaching and Academic Services

**Teaching Experience**: TA for Georgia Tech CSE-8803 Statistical Machine Learning Models for Neural and Behavioral Data Analysis.

**Conference Reviewer**: NeurIPS 2023/2024/2025/2026, ICML 2024/2025/2026, ICLR 2024/2025/2026.